Redshift Data Access Request

Plivo Log Analysis Dashboard — Field-Level Justification

Requested by: Suraj Menon **Date:** March 9, 2026 **Purpose:** Read-only access to specific columns for the Plivo Log Analysis Dashboard, an internal tool that automates customer issue diagnosis for support agents. **Access Mode:** SELECT only. No INSERT, UPDATE, DELETE, CREATE, or DDL operations. **Row Limit:** All queries will be capped at 150 rows per request to minimize scan cost.

How the Dashboard Uses This Data

The tool follows a three-step pipeline:

1. **Fetch** — Query Redshift with customer identifiers (account ID or username) and a date range. Return up to 150 matching log records.
2. **Analyze** — Aggregate the records to compute error distributions, success rates over time, carrier/region breakdowns, and health scores. Feed aggregated stats to an AI model for pattern detection and recommendations.
3. **Present** — Display interactive charts, a filterable log table, trend indicators, and a downloadable PDF report to the support agent.

Each field requested below maps directly to one or more of these steps. Fields are grouped by table with a clear justification for each.

Table 1: `base.mdr_raw_airflow`

Purpose: Primary source for SMS/MMS (MDR) log analysis. **Why this table over `fact_mdr_enriched`:** Contains full phone numbers (not redacted), human-readable `carrier_name`, and `username` for customer lookup — all essential for support agent workflows.

#	Column	Type	Justification
1	`message_uuid`	varchar(64)	**Unique record identifier.** Used as the primary key in the log table. Allows agents to reference specific messages in Zendesk tickets and cross-reference with other internal tools.
2	`account_id`	varchar(16)	**Customer lookup.** Primary filter for querying logs by Plivo Auth ID. This is how agents identify which customer's logs to analyze.
3	`username`	varchar(32)	**Customer lookup (alternate).** Agents sometimes only have a customer email/username. This allows filtering by that identifier when Auth ID is unavailable.
4	`from_number`	varchar(100)	**Source number display.** Shown in the log table so agents can identify the sending number. Helps diagnose number-specific issues (e.g., a particular sender ID being blocked by carriers).
5	`to_number`	varchar(100)	**Destination number display.** Shown in the log table. Essential for identifying destination-specific delivery failures (e.g., invalid number format, country-specific blocking).
6	`message_direction`	varchar(16)	**Traffic direction filter.** Values: `inbound`/`outbound`. Allows agents to filter and analyze inbound vs outbound traffic separately, as failure patterns differ by direction.
7	`message_state`	varchar(16)	**Success/failure classification.** Values: `delivered`, `undelivered`, `sent`, etc. This is the primary field for computing success rates, health grades, and

			identifying failed records for error analysis.
8	`message_type`	varchar(8)	**Message type filter.** Values: `sms`/`mms`. Allows agents to isolate SMS vs MMS issues, which have different carrier routing and failure modes.
9	`dlr_error`	varchar(16)	**Error code for failure analysis.** Values: `000` (success), `801`, `300`, `006`, etc. This is the primary field for building the error distribution chart and classifying failure types. Maps to specific delivery failure reasons.
10	`carrier_name`	varchar(64)	**Carrier attribution.** Values: `clx`, `mitto-standard`, `sap-33433`, etc. Used to build carrier distribution charts and identify carrier-specific failure patterns (e.g., "80% of failures are on carrier X"). Critical for actionable recommendations.
11	`carrier_id`	varchar(16)	**Carrier identifier.** Used alongside `carrier_name` for precise carrier identification when carrier names are ambiguous or when cross-referencing with internal carrier configuration.
12	`country_iso`	varchar(4)	**Country/region analysis.** Values: `US`, `AU`, `IR`, `JP`, etc. Used to build geographic distribution charts, identify country-specific delivery issues, and filter logs by destination country.
13	`to_iso`	varchar(8)	**Destination country code.** Used as a secondary geographic classifier. In some records, `to_iso` may differ from `country_iso` for routing analysis.
14	`units`	integer	**Message segment count.** Indicates how many SMS segments a message was split into. Relevant for cost analysis and identifying issues with long messages that get split and partially fail.
15	`message_time`	timestamp	**Timestamp for time-series analysis.** Used to plot success rate over time, group records by date, detect trends (improving/worsening), and filter logs within the requested date range.
16	`number_type`	varchar(32)	**Number type classification.** Values: `local`, `mobile`, `shortcode`, `tollfree`. Allows agents to filter and identify if failures are concentrated on a specific number type (e.g., shortcode messages failing more than long codes).

Table 2: `base.fact_mdr_enriched`

Purpose: Secondary/enriched source for SMS analysis when additional error classification is needed. **When used:** As a supplementary lookup when `dlr_error` codes from `mdr_raw_airflow` need enriched error categorization.

#	Column	Type	Justification
1	`message_uuid`	varchar(36)	**Join key.** Used to correlate records between `mdr_raw_airflow` and this enriched table when additional error context is needed.
2	`account_id`	integer	**Customer filter.** Used when querying this table directly for enriched error data.
3	`message_state`	varchar(12)	**Status verification.** Cross-reference delivery status with raw table to confirm consistency.
4	`dlr_error`	varchar(10)	**Error code (enriched).** Same DLR error code, used for joining and verification.
5	`enr_error_code`	integer	**Enriched error classification.** Provides a normalized, integer-based error code that maps to standardized error categories. Supplements the raw

			`dlr_error` string with a cleaner classification for AI analysis.
6	`carrier_id`	varchar(28)	**Carrier identification.** Since this table lacks `carrier_name`, the ID is used to join back to raw data or internal carrier mapping for attribution.
7	`country_iso`	varchar(4)	**Geographic analysis.** Same purpose as in raw table — country-level filtering and distribution charts.
8	`message_time`	timestamp	**Temporal alignment.** Used for date-range filtering when querying this table.
9	`dst_number_type`	char(2)	**Enriched number type.** Values: `LC` (local), `SC` (shortcode), `TF` (tollfree), `MB` (mobile). Provides standardized number type classification for more consistent analysis.
10	`message_direction`	varchar(8)	**Direction filter.** Same purpose as in raw table — filtering by inbound/outbound traffic.

Table 3: `base.cdr_raw_airflow`

Purpose: Primary source for Voice (CDR) log analysis. **Why this table:** Contains the richest set of call detail fields including Plivo-specific hangup codes, carrier names, call quality indicators, and customer identifiers.

#	Column	Type	Justification
1	`call_uuid`	varchar(64)	**Unique record identifier.** Primary key for the log table. Allows agents to reference specific calls in tickets and trace call flows.
2	`account_id`	varchar(16)	**Customer lookup.** Primary filter for querying logs by Plivo Auth ID.
3	`username`	varchar(32)	**Customer lookup (alternate).** Values: `schoolpointe`, etc. Allows filtering by customer username when Auth ID is unavailable.
4	`from_number`	varchar(128)	**Caller number.** Displayed in log table. Helps identify caller-specific issues (e.g., a number being flagged as spam).
5	`to_number`	varchar(128)	**Destination number.** Displayed in log table. Essential for diagnosing destination-specific routing failures.
6	`call_direction`	varchar(8)	**Traffic direction.** Values: `inbound`/`outbound`. Filters and segments analysis by direction, as failure patterns differ significantly.
7	`call_state`	varchar(32)	**Call outcome.** Values: `ANSWER`, `NOANSWER`, `BUSY`, `CANCEL`, etc. Used alongside hangup codes to classify call success/failure.
8	`hangup_cause`	varchar(32)	**Hangup reason (SIP-level).** Values: `NORMAL_CLEARING`, `USER_BUSY`, `NO_ROUTE_DESTINATION`, etc. Primary field for the error distribution chart. Maps to standard SIP/Q.850 cause codes.
9	`plivo_hangup_cause_code`	integer	**Plivo-specific hangup code.** Values: `4000` (Normal), `4010` (End of XML), etc. Numeric code used for programmatic success/failure classification. Codes 4000 and 4016 indicate success.
10	`plivo_hangup_cause_name`	varchar(64)	**Plivo-specific hangup description.** Values: `Normal Hangup`, `End Of XML Instructions`, etc. Human-readable label shown in the error distribution chart and issue cards. More descriptive than raw SIP causes.

11	`plivo_hangup_source`	varchar(16)	**Who initiated the hangup.** Values: `Callee`, `Caller`, `Plivo`, `Carrier`. Helps agents determine whether failures originate from the customer's app, the end user, the carrier, or Plivo infrastructure.
12	`carrier_name`	varchar(64)	**Carrier attribution.** Values: `Level_3_Communication`, etc. Used to build carrier distribution charts and identify carrier-specific routing failures.
13	`carrier_id`	varchar(32)	**Carrier identifier.** Precise carrier reference for cross-referencing with internal carrier dashboards.
14	`country_iso`	varchar(4)	**Destination country.** Values: `US`, `UK`, `IN`, etc. Used for geographic distribution analysis and country filtering.
15	`from_iso`	varchar(5)	**Origin country.** Used alongside `to_iso` for international call routing analysis.
16	`to_iso`	varchar(8)	**Destination country code.** Alternate geographic classifier for destination.
17	`start_time`	timestamp	**Call initiation timestamp.** Used for time-series charts, date-range filtering, and trend detection.
18	`answer_time`	timestamp	**Call answer timestamp.** Used to calculate ring time and post-dial delay. Null values indicate unanswered calls.
19	`end_time`	timestamp	**Call end timestamp.** Used with `start_time` to compute actual duration and for date-range boundary filtering.
20	`bill_duration`	integer	**Billable duration (seconds).** Shown in log table. Helps agents assess call impact — zero-duration calls with failure codes indicate complete routing failures vs. calls that connected but dropped.
21	`ring_time`	integer	**Ring duration (seconds).** Diagnostic indicator. Excessively long ring times may indicate destination issues. Zero ring time with failure may indicate immediate rejection.
22	`post_dial_delay`	integer	**Post-dial delay (seconds).** Network quality indicator. High PDD values may indicate carrier-side latency or routing issues. Used in AI analysis for quality assessment.
23	`sip_call_id`	varchar(128)	**SIP Call-ID header.** Used by agents for deep debugging — can be shared with carriers to trace specific calls through their infrastructure.
24	`tollfree`	varchar(8)	**Toll-free indicator.** Values: `True`/`False`. Allows filtering by toll-free traffic, which has distinct carrier routing and failure patterns.
25	`call_duration_type`	varchar(16)	**Duration bucket.** Values: `30S`, `60S`, etc. Helps segment analysis by call length to identify if short/long calls fail at different rates.

Table 4: `base.zentrunk_cdr_raw`

Purpose: Primary source for Zentrunk (SIP Trunking) log analysis. **Why this table:** Zentrunk operates differently from standard Plivo voice — it's a SIP trunking product with different fields (trunk-level routing, SIP codecs, transport protocols).

#	Column	Type	Justification

1	`call_uuid`	varchar(100)	**Unique record identifier.** Primary key for Zentrunk calls in the log table.
2	`account_id`	integer	**Customer lookup.** Primary filter for querying Zentrunk logs by customer account.
3	`from_number`	varchar(256)	**Source identifier.** Values include SIP usernames (e.g., `mathrawkdev`) and phone numbers. Displayed in log table for call identification.
4	`to_number`	varchar(256)	**Destination number.** Displayed in log table for diagnosing destination-specific routing issues.
5	`call_direction`	varchar(20)	**Traffic direction.** Values: `inbound`/`outbound`. Segments analysis by direction.
6	`hangup_cause`	varchar(100)	**Hangup reason.** Values: `normal_clearing`, `originator_cancel`, `no_route_destination`, etc. Primary field for error distribution chart and failure classification.
7	`hangup_code`	integer	**Numeric hangup code.** SIP/Q.850 cause code for programmatic classification. Used to determine success (null or normal codes) vs failure.
8	`hangup_initiator`	varchar(100)	**Who terminated the call.** Values: `customer`, `carrier`, `callee`. Crucial for Zentrunk diagnosis — determines whether failures originate from the customer's PBX, the carrier, or the far end.
9	`carrier_id`	varchar(16)	**Carrier identifier.** Used for carrier-level failure attribution and distribution charts.
10	`carrier_gateway`	varchar(128)	**Carrier SIP gateway.** Values: `sip:+18593618614@4.55.40.227`. Contains the actual SIP URI used for routing. Helps diagnose gateway-specific failures (e.g., a particular gateway IP rejecting calls).
11	`from_iso`	varchar(5)	**Origin country.** Geographic analysis of call origination.
12	`to_iso`	varchar(16)	**Destination country.** Geographic analysis for regional failure patterns.
13	`initiation_time`	timestamp	**Call start timestamp.** Primary timestamp for time-series analysis, date-range filtering, and trend detection.
14	`answer_time`	timestamp	**Call answer timestamp.** `1970-01-01` indicates unanswered calls. Used to distinguish between routing failures and answered-then-dropped calls.
15	`end_time`	timestamp	**Call end timestamp.** Used with `initiation_time` for duration calculation and date-range boundaries.
16	`duration`	integer	**Actual call duration (seconds).** Zero indicates failed/unanswered calls. Shown in log table.
17	`bill_duration`	integer	**Billable duration (seconds).** Shown alongside actual duration. Helps agents understand billing impact of failed calls.
18	`transport_protocol`	varchar(25)	**SIP transport.** Values: `udp`, `tcp`, `tls`. Zentrunk-specific diagnostic — transport choice can affect reliability. Allows filtering by protocol if failures correlate with a specific transport.
19	`srtp`	boolean	**Media encryption indicator.** Whether SRTP was used. Relevant for diagnosing media-related failures where encryption negotiation fails.
20	`src_codec`	varchar(512)	**Source codec list.** Values: `PCMU,telephone-event/8000,CN`. Diagnostic field for media quality issues — codec mismatches between source and

			destination can cause call failures or quality degradation.
21	`dest_codec`	varchar(128)	**Destination codec list.** Compared with `src_codec` to identify codec negotiation issues.
22	`tollfree`	varchar(8)	**Toll-free indicator.** Allows filtering toll-free trunk traffic separately.

Summary

Table	Fields Requested	Total Columns in Table	Access Scope
`base.mdr_raw_airflow`	16 of 39	39	SELECT only, WHERE on `account_id`/`username` + `message_time`, LIMIT 150
`base.fact_mdr_enriched`	10 of 31	31	SELECT only, WHERE on `account_id` + `message_time`, LIMIT 150
`base.cdr_raw_airflow`	25 of 56	56	SELECT only, WHERE on `account_id`/`username` + `start_time`, LIMIT 150
`base.zentrunk_cdr_raw`	22 of 30	30	SELECT only, WHERE on `account_id` + `initiation_time`, LIMIT 150

What We Are NOT Accessing

The following sensitive/irrelevant columns are explicitly **excluded** from all queries:

- **Billing/financial data:** `cloud_rate`, `carrier_rate`, `total_cloud_amount`, `total_carrier_amount`, `carrier_cost`, `total_amount`, `rate`, `enr_cost`, `termination_rate` — not relevant to log diagnosis
- **Account configuration:** `account_type`, `auto_payment_approval`, `product_plan`, `tier`, `tier_name`, `plan_id` — internal billing metadata
- **Infrastructure internals:** `server_id`, `server_ip`, `opensips_ip`, `extra_data`, `conversion_rate` — internal infrastructure details not needed for customer-facing analysis
- **Internal IDs:** `id`, `subaccount_id`, `sub_account_id`, `parent_call_uuid`, `request_uuid`, `parent_uuid`, `parent_message_uuid`, `message_part_string`, `sms_group_id`, `product_plan_id`, `billing_tier_id` — internal references not useful for support diagnosis
- **Redundant fields:** `rounded_bill_duration` (have `bill_duration`), `conference_uuid` (not in scope), `last_updated`, `message_charge_time`

Security Controls

1. **Read-only access** — All queries are `SELECT` statements only. No write operations of any kind.
2. **Row limit** — Every query includes `LIMIT 150` to cap data scanned per request.
3. **Scoped filters** — All queries filter by `account_id` or `username` + date range. No full-table scans.
4. **No data persistence** — Fetched records are held in browser `sessionStorage` only and auto-clear after 15 minutes of inactivity.
5. **No PII export to external services** — Raw log data stays client-side. Only aggregated statistics (counts, percentages) are sent to the AI model for analysis.